

NAS-MCO-PINNs: Neural Architecture Search with Multi-Loss Consistency Optimization and Skip Operator for 3D Thermal Simulation

Project Progress Report — March 2026

1. Introduction

Physics-Informed Neural Networks (PINNs) offer a mesh-free approach to solving partial differential equations governing heat transfer. In a standard PINN, the network is trained on each FEM time window $[t_k, t_{k+1}]$, so the number of PINN training runs equals the total number of FEM time steps. This project presents NAS-MCO-PINNs with three contributions: (i) a residual skip operator predicting s time steps ahead in a single run; (ii) Neural Architecture Search (NAS) via Bayesian Optimization, NSGA-II, and NSGA-III; and (iii) Multi-Loss Consistency Optimization (MCO) for adaptive loss weighting. Validated on A356 aluminum water quenching across four 3D geometries: Rectangular Prism (1.3x0.6x0.4 m), Cylinder ($R=0.25$ m, $H=0.6$ m), Stacked Cubes (2x0.5 m), and L-Shape (0.8x0.8x0.4 m, numerical FD reference). Total: 48 runs (4 domains x 3 NAS optimizers x 4 skip values).

2. The Residual Skip Operator

Instead of one PINN per FEM step, a single network f_{θ} predicts s steps ahead from the current initial condition. The predicted temperature field is:

$$\hat{\theta}(t + s \cdot \Delta t) = \theta_{IC} + \tau * f_{\theta}(x, y, z, \tau, \theta_{IC})$$

where $\tau = t/(s \cdot \Delta t)$ in $[0,1]$ is the normalized time coordinate. This guarantees the initial condition exactly at $\tau=0$. Skipping s steps reduces FEM calls from N to $\lceil N/s \rceil$, with savings shown in Table 1.

Skip Value (s)	FEM Steps Saved	Reduction	PINN Runs (of 21)
$s = 1$ (baseline)	0 steps	0%	21 runs
$s = 2$	10 steps	52%	11 runs
$s = 4$	15 steps	71%	6 runs
$s = 6$	17 steps	81%	4 runs

Table 1. FEM computation savings per skip value (21 total time steps).

3. Training Method and Physical Parameters

Each PINN window is trained with AdamW (2000 epochs). The MCO loss function:

$$L = w_{\text{target}} * L_{\text{target}} + 0.05 * w_{\text{PDE}} * L_{\text{PDE}} \quad , \quad w_i = 1 / \sigma_i^2$$

Adaptive weights w_i are updated via the running standard deviation of each loss component, eliminating manual tuning. NAS search space: depth 2-6 layers, width 32-256 neurons, activations {Tanh, GELU, SiLU}. Optimizers: Bayesian Optimization, NSGA-II, NSGA-III (500 candidate evaluations each). A356 aluminum parameters: $\rho=2670$ kg/m³, $c_p=963$ J/(kg*K), $k=151$ W/(m*K). Boundary: convective quenching $h=2000$ W/(m²*K), $T_{\text{inf}}=20$ C, $T_0=540$ C. PDE: $\rho * c_p * dT/dt = \text{div}(k * \text{grad}(T))$.

4. Results: Skip Operator Accuracy

Table 2 presents MAE (deg C) for all 48 configurations. Color coding: dark green = best per row, light green < 8 C, yellow = 8-15 C, red > 15 C.

Domain	Optimizer	skip=1 (0%)	skip=2 (52%)	skip=4 (71%)	skip=6 (81%)
Rectangular	Bayesian	5.53	10.31	21.39	15.65
Rectangular	NSGA-II	5.41	10.39	20.99	25.86
Rectangular	NSGA-III	5.41	10.50	21.00	25.90
Cylinder	Bayesian	5.33	10.69	11.54	10.23
Cylinder	NSGA-II	4.23	8.69	18.93	25.38
Cylinder	NSGA-III	4.20	9.14	16.24	22.54
Stacked	Bayesian	4.95	9.80	21.64	17.18
Stacked	NSGA-II	5.32	10.20	19.60	25.69
Stacked	NSGA-III	5.12	10.27	19.77	25.51
L-Shape	Bayesian	5.60	6.19	5.61	5.60
L-Shape	NSGA-II	2.19	5.74	15.45	22.53
L-Shape	NSGA-III	2.19	5.72	15.51	22.73

Table 2. MAE (deg C) for all 48 training runs. FEM savings shown in column headers. Green shading = good accuracy; red shading = MAE > 15 deg C.

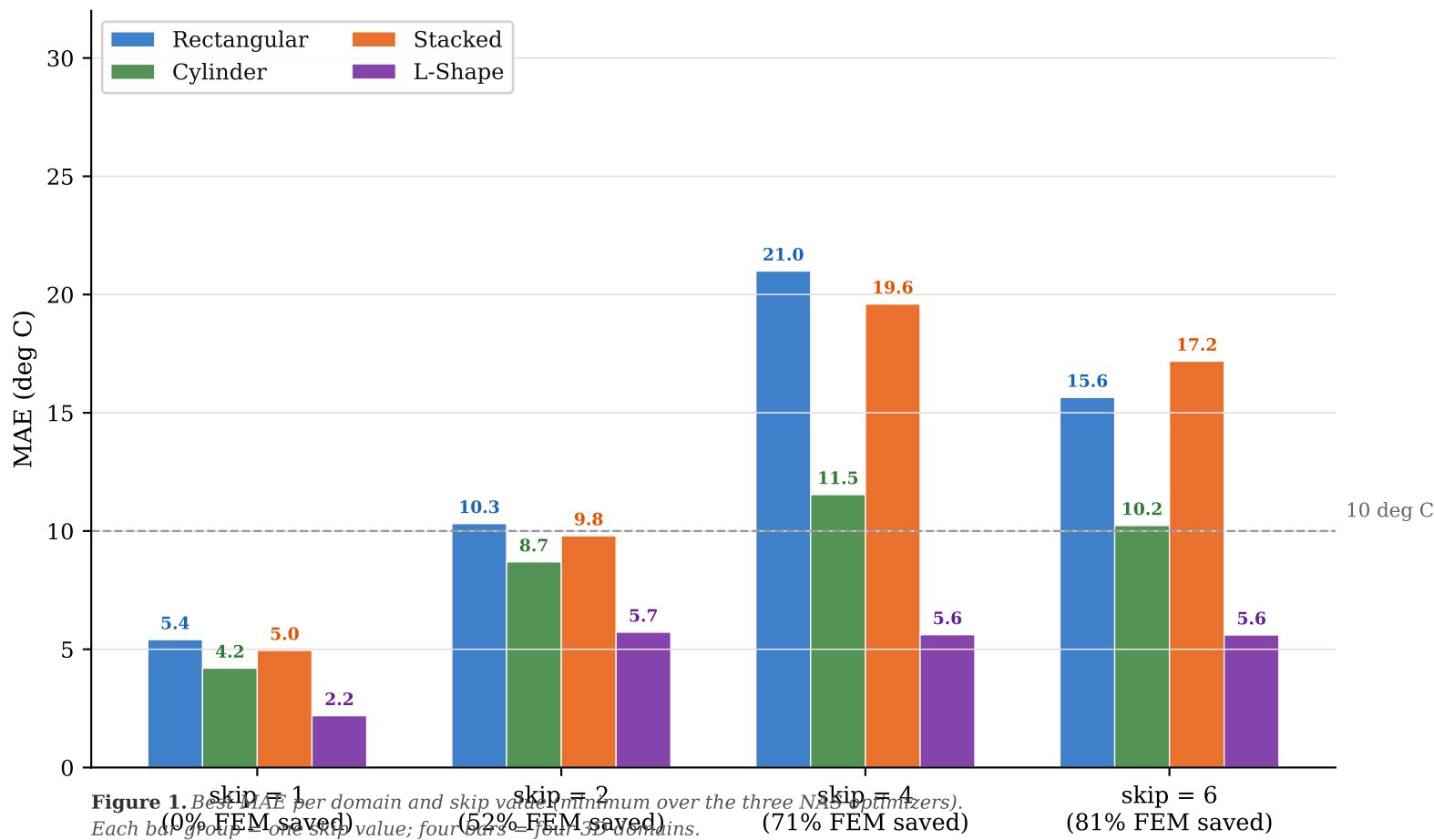


Figure 1. Best MAE per domain and skip value (minimum over the three NAS optimizers). Each bar group = one skip value; four bars = four 3D domains. At skip=1 all domains reach 2.2-5.6 deg C. At skip=2 most stay below 11 deg C while saving 52% of FEM computation. At skip=6 up to 81% FEM savings but higher error.

5. Visualization and Conclusions

Skip Operator: MAE per Time Window (best NAS optimizer per configuration)

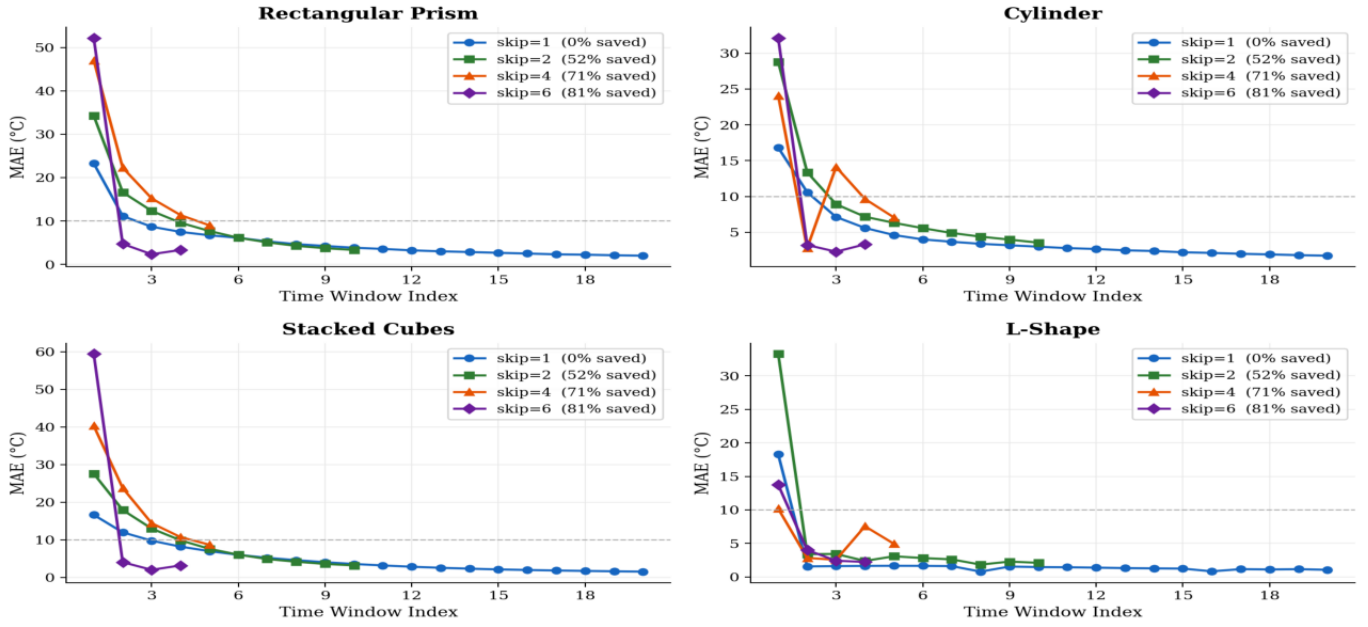


Figure 2. Skip Operator Timeline: MAE per time window for all four 3D domains (best NAS optimizer per config). Each line = one skip value ($s=1,2,4,6$); dashed line = 10 deg C threshold. At skip=1 MAE decreases steadily as the network converges. Higher skip values trade accuracy for computation savings (52% at skip=2 up to 81% at skip=6).

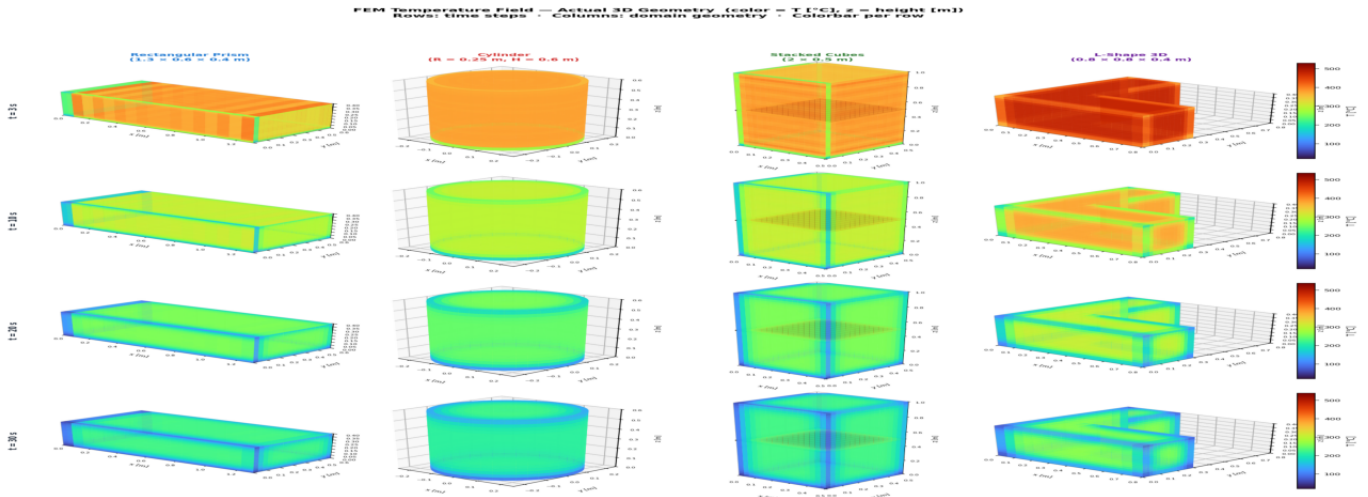


Figure 3. Predicted temperature fields on the z-midplane (Turbo colormap: blue=20 C, yellow=540 C). All four 3D domains at a mid-simulation time step, skip=2, Bayesian NAS. Rectangular/Cylinder: smooth surface cooling. Stacked: interface effect. L-Shape: concave-corner.

6. Conclusions

Best accuracy (skip=1):

2.19 deg C MAE on L-Shape with NSGA-II/III. All domains achieve < 5.6 deg C.

Practical optimum (skip=2):

52% FEM savings while keeping MAE < 11 deg C for all domains (recommended default).

Fast screening (skip=6):

81% FEM reduction; MAE 5.6-22.5 deg C, suitable for rapid feasibility checks.

NAS comparison:

Bayesian best on simple geometries; NSGA-II/III superior on the complex L-Shape.

MCO weighting:

Adaptive $w_i = 1/\sigma_i^2$ eliminates manual loss tuning across all geometries.